Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Topi, Swabi

Faculty of Computer Science and Engineering

Ink-Finity

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Certificate of Approval

It is certified that the work presented in this report was performed by **Alishba Faheem, Waqar Babar, Syed Ahmad Mansoor and Najeed Mubasher** under the supervision of **Dr. Rashid Jillani.** The work is adequate and lies within the scope of the BS degree in Computer Science/Computer Engineering at Ghulam Ishaq Khan Institute of Engineering Sciences and Technology.

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(Dean)

SDGs Inclusion

This project directly supports three major Sustainable Development Goals (SDGs):

SDG 4 – Quality Education: The project increases learning effectiveness and accessibility, particularly for visual and auditory learners, by using generative AI to create captivating, personalized educational videos and shorts.

SDG 8 – Decent Work and Economic Growth: The platform empowers creators worldwide by facilitating scalable content creation and creating opportunities for micro-entrepreneurship in digital content production, animation, and voice-over services.

SDG 9 – Industry, Innovation, and Infrastructure: The future of creative AI industries is supported by the integration of advanced AI pipelines (text-to-3D, text-to-video, and shorts generation), which encourage innovation in digital media infrastructure.

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Dr. Rashid Jillani	**Dr. Sara Iqbal**
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FINAL YEAR PROJECT MAPPING TO

COMPLEX ENGINEERING/COMPUTING PROBLEM ATTRIBUTES

It is to certify here that the final year design project (FYDP) entitled Ink-Finity is categorized as a complex engineering/computing problem (CEP) based on the preamble (in-depth engineering knowledge) and involvement of the following attributes:

Attribute	**Justification**
Range of Conflicting Requirements	Balancing inference speed with output quality; high-resolution outputs vs. real-time generation constraints
Depth of Analysis Required	Required architectural tuning, prompt engineering, model evaluation (e.g CLIP score, Chamfer Distance).
Depth of Knowledge Required	Involves deep learning, computer vision, NLP, audio synthesis and pipeline integration
Familiarity of Issues	New and already practiced AI tools and models were used so some problems faced were unfamiliar while others were familiar.
Extent of Applicable Codes	Involves open-source AI frameworks, content policy constraints (e.g for TTS, generative safety)
Extent of Stakeholder Involvement and Level of Conflicting Requirements	Users demand fast, engaging output; developers prioritize accuracy, safety and ethical use of AI
Consequences	Misinformation risk, biased generation, or inappropriate outputs can reduce user trust and usability.
Interdependence	Image, audio and text modules are tightly coupled. Failure in one affects the entire pipeline

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Dr. Rashid Jillani **Dr. Sara Iqbal**

(Advisor) (Co-Advisor)

ABSTRACT

Ink-Finity is a system that aims to simplify the creation of comic panels and animated videos by utilizing AI technology. It allows educators, marketers, content creators, and even hobbyists to generate 2D and 3D animations with ease, using only text prompts, and without needing any technical skills. Ink-Finity combines natural language processing, generative AI models, and user-friendly web applications to bring users' ideas to life in high-quality downloadable formats. Compared to traditional methods of animation, Ink-Finity is creative, quick, easy to use, and sustainable in terms of reliability and ethics.

ACKNOWLEDGEMENTS

We thank our advisors, both **Dr. Rashid Jillani** and **Dr. Sara Iqbal**, for providing us with the utmost support and guidance. They played a critical role in the success of our efforts. We thank the faculty and the institute for providing us with the best educational opportunities. This project would not be possible without the help of various other professors, and researchers across the globe, who paved the way for us to be able to create this project.

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CHAPTER I

1.INTRODUCTION

1.1 PURPOSE

We now experience how everything has shifted to solutions generated through artificial intelligence. Online, traditional animation is time-consuming and requires artistic and software skills. Statistically, demand for animation is projected to grow at a compound annual growth of over 5% globally, hence, making it a growing market of interest. Ink-Finity is going to be a successful product in this market to streamline this process, allowing users to produce short and high-quality animated videos at ease.

1.2 PRODUCT SCOPE

This product is designed to cater to a diverse group of users who are interested in creating animated content but lack the technical skills and resources typically required. The main users will be content creators, such as people who are creating animated content on YouTube and Instagram. We often see voice over animated videos on YouTube hence, social media users are a target audience. Secondly, educators and students both are a target audience. Educators can benefit from the product by creating animated videos to simply complex contents for students.

The key features and scope of the product include:

1.2.1 GENERATING COMIC PANELS

The system will take descriptions of the characters, settings, and scenarios from the user and pass them to an AI model that will use the text to generate comic panels. The system will offer the user the option to download the generated in any of the preferred file formats (e.g. PNG, JPEG)

1.2.2 GENERATING 2D VIDEO

The system generates 2D videos from the given user input. The comic generated and the text input provided work as the input for the video generation model. A fine-tuned StabilityAI is the proposed model for video generation.

1.2.3 GENERATING 3D VIDEO

The system will take the previous inputs and generate a 3D video of the user-defined setting and characters. The 3D video will be 5-10 seconds in length and will be dependent on user's choice.

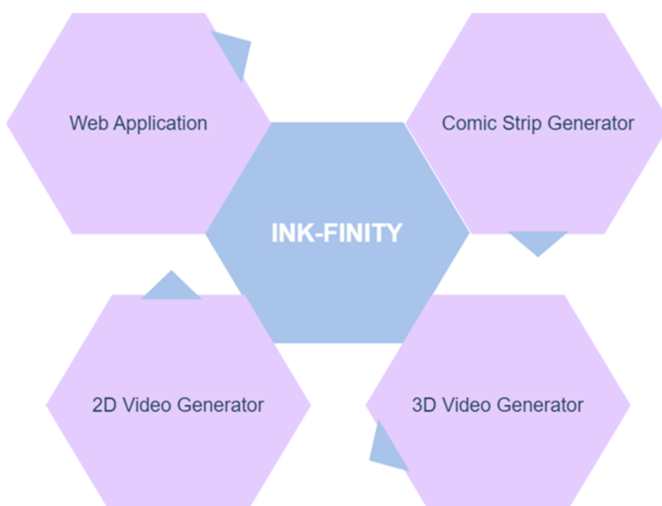

Figure 1: Workflow

1.3 OVERVIEW

1.3.1 OVERALL DESCRIPTION

This document outlines the design, architecture, and core functionalities of ink-Finity, an AI-powered animation generation system. It is intended to simplify the creation of animated content through a web-based platform where users can input the description of the graphic to initiate the animation generation process. The primary goal is to enable users without technical and artistic backgrounds to benefit from their imagination to create a product worthy of appreciation.

Ink-Finity features a web application frontend that captures the user's input, the input is redefined using Natural language processing techniques to be useful for the AI model. The input will be pre-processed to convert from raw data into a structured form using NLP EDA and parsing techniques. The input is then passed to an AI model which will use the text description containing character stories, setting and scenarios to be generated to make comic panels to be displayed to the user. The user will have the ability to make edits and re-generate the output. The comic is then in return passed to the 2D video generation model which uses the inputs to convert a scene from the comic into a video which will be available to download as an mp4 animated. The last feature of the AI model will be to convert the 2D video into a 3D video snippet upon user's request. The system will have multiple sketches as PNG and JPEG images (based on the option selected by the user) as an output to the frontend which is downloadable.

1.3.2 PRODUCT PERSPECTIVE

The core objective of this project is to reduce the time required, generating animated content from Photoshop or other software generally consumes hours of hard work which will be reduced to a fraction of a second leading to an improved performance. It should be able to generate high-quality videos which are fulfilling the provided user requirements.

The product is aimed at making creativity and art available to all users from any background. The current tools in the market require expert knowledge but this product aims to bridge the gap between imagination and implementation. It debunks all stereotypes of AI killing imagination in humans and leverages the powers of AI to simplify the complex process of creating comics.

1.3.3 PRODUCT FUNCTIONS

Ink-Finity an AI-based animation generation system offers several key functionalities to streamline the process of animated video generation through user input for educational and industrial purposes:

1.3.4 DESCRIPTION TEXT FUNCTION

The function handles providing user with dialogue box to give input for the model. This input will include details about the characters, settings of the comic and the scenario the comic is based on along with any other details they wish to provide. The Web Application will forward this information to the AI model that will be linked via APIs. The user will also be able to choose animation style from a list of provided popular styles (e.g. Manga).

1.3.5 PREPROCESSING FUNCTION

The AI Model, upon receiving the raw text will perform pre-processing on the text. This will involve structuring the data into the required fields of characters, character descriptions, setting, scenario and dialogues(optional). The text will also be cleaned during this step and converted to vectors to be passed to the Comic-generator module.

1.3.6 COMIC GENERATION MODULE

The structured and processed text input will be given to the AI Comic generator module. The module will use fine-tuned StabilityAI and custom-built LSTMs and GANs to generate the Comic panels. The number of panels per comic will be limited to 4 and hard coded instead of giving user the ability to choose. The output from this module will be a .png or .jpeg file that will be passed on to the Web Application and displayed to the user.

1.3.7 VIDEO GENERATION MODULE (2D AND 3D)

The system will pass the generated comic and the text input to Video generator models. These models will generate short snippets from the given comic scenario ranging from 5-10 seconds.

1.3.8 WEB APPLICATION

A user-friendly web application that will be used as the front-end for the AI models to make it accessible for all expertise levels. It would use Next.JS as the front-end and Flask & NodeJS will be used for the backend. The backend will integrate the AI model using APIs.

1.3.9 USER CHARACTERISTICS

The system is designed for a variety of user base, spanning from user with non-technical backgrounds to those with moderate level of technical expertise. While specific animation skills are not a prerequisite, the primary users are expected to possess familiarity with general web applications and digital content creation ideas. Users are primarily motivated by the need to generate engaging and attractive visual content within seconds, without the complexities of traditional methods which require extensive knowledge of softwares such as adobe photoshop and premiere pro.

1.3.9.1 CONTENT CREATORS

These users rely on accessible tools to create short but visually appealing content for the use of their social media platforms such as YouTube. Their primary purpose is to convey the idea using visually appealing content to help them maintain their engagement with their audience.

1.3.9.2 SMALL BUSINESS OWNERS AND MARKETERS

This category of users is based mostly on startup owners who have to invest heavily to get customer engagement for their businesses, this will provide them with an affordable solution to market their products.

1.3.9.3 EDUCATORS AND E-LEARNING PLATFORMS

In education and digital learning environments, animation serves as a valuable tool to simply the concepts and engage students along with being a cost-effective substitute for physical instructors. This helps to create easy content that can result in maximum student engagement (Such as simplilearn platform).

1.3.9.4 AMATEUR ANIMATORS AND HOBBYISTS

Individuals in this group are primarily driven by their visual and creative explorations, however, the lack of video generation skills and unfamiliarity with the software act as a hurdle to their creative freedom.

1.3.10 CONSTRAINTS

The development and deployment of the ink-Finity animation generation system must consider several technical, operational, and resource-based constraints:

1.3.10.1 COMPUTATIONAL RESOURCE REQUIREMENTS

Generating high-quality animated frames from text descriptions demand significant computation power when scaling to support simultaneous users at a certain time. Limited GPU resources might affect the processing speed and might impact user experience.

1.3.10.2 RESPONSE TIME

The target is to produce downloadable content in real-time, however, maintaining a low latency can be challenging due to the complexities of AI-driven frame generation and video synthesis, which may require further optimization to meet performance requirements.

1.3.10.3 LIMITATIONS OF MODELS

The quality of the output is based on the training of the model on the data set. This will require extensive training and pre-processing, however, still this constraint may limit the system’s ability to handle detailed and nuanced animation requests beyond the general themes or the pre-trained themes within the model.

1.3.10.4 ETHICAL CONSTRAINTS

The AI model can be used irresponsibly to generate ethically problematic content. Future directions of this project will include adding ethical constraints to avoid such situations.

1.3.11 ASSUMPTIONS AND DEPENDENCIES

The successful implementation and functionality of the system depends on specific assumptions and external dependencies regarding performance, usability and operational stability:

1.3.11.1 USER INPUT RELEVANCE

It is assumed that the user provided text inputs that are clear and provide a suitable context for animation generation with essential details regarding the scenes and characters to be included in the sketches or the animated video.

1.3.11.2 MODEL TRAINING

The accuracy and creativity in the generated content is dependent on a well-trained AI model. For that reason, the performance of the system is highly dependent on the availability of extensive dataset which is not tailored to a certain common theme for video generations.

1.3.11.3 PROCESSING CAPABILITIES

It assumed that there are adequate GPU resources to maintain the performance and efficiency of the model.

1.3.11.4 DEVICE COMPATIBILITY

System assumes that the users will access the platform through browsers that are compatible with the system requirements to provide a smooth user experience.

CHAPTER II

2.LITERATURE SURVEY

2.LITERATURE REVIEW

2.1 LANGUAGE-GUIDED FACE ANIMATION BY RECURRENT STYLEGAN-BASED GENERATOR

2.1.1 METHODOLOGY

This study addresses the novel task of language guided face animation, building upon existing models in place for language guided image manipulation. While the prior research has demonstrated the incorporation of motion dynamics in language input to animate static face images. The method in this research paper leverages both semantic and motion description extracted from the language input to guide the animation.

The authors of this research introduce a framework based on the recurrent motion generator that processes language descriptions after extracting sequential semantic and motion details to the animate a still face in a frame. The data provided is then passed and processed by a pretrained StyleGAN to produce realistic and animated frames. The framework uses three designed loss function: a regularization loss, a path length regularization loss and contrastive loss function to increase smoothness in the video.

2.1.2 FINDINGS

The results from evaluation demonstrate the model's effectiveness in generating high quality, realistic face animations across the domains. The results reaffirm that this recurrent StyleGAN based framework enables a more seamless and adaptable approach to video synthesis driven by language.

2.1.3 LIMITATIONS

Despite the performance achieved by the model it is highly dependent on high quality language descriptions for accurate and real animations. Variability in language interception may affect the model's ability to generate coherent and accurate animations as requested by the user.

2.2 FREE-BLOOM: ZERO-SHOT TEXT-TO-VIDEO GENERATOR WITH LLM DIRECTOR AND LDM ANIMATOR

2.2.1 METHODOLOGY

This paper presents the free-bloom framework an approach to generate zero shot text to video. The objective of this research is to allow the creation of high-quality ideas solely from text prompts without relying on video datasets or requiring extensive training. Moreover, it uses large language models to parse sequential prompts and latent diffusion models to animate each frame achieving temporally coherent and semantically aligned video sequences.

2.2.2 FINDINGS

Free bloom maintains temporal consistency across multiple frames while accurately depicting the provided scenes in the input text. It achieves high fidelity and coherence, effectively capturing motions dynamics and transitions.

Free bloom model was compared to other baseline models which includes LVDM and T2V-Zero and revealed that free-bloom surpasses other methods in generating frames with smoother transitions and adherence to input, rating 4.122 for fidelity and 3.867 for semantic coherence.

It is highly effective in depicting complex scenarios such as teddy bear jumping into water and volcano erupting showcasing its ability to create semantically meaningful content.

2.2.3 LIMITATIONS

This approach is highly dependent on pre-trained LLM's and LDM's, making it susceptible to the limitations of those baseline models. Complex scenes and inputs on which the model is not trained might result in incorrect depiction of the scenes.

The framework is highly sensitive to noise; hence, extensive data preprocessing and noise reduction is required.

The framework is effective in producing shorter videos however, there are inconsistent frames and experiences difficulties in coherence for scenes of longer durations.

2.3 MOTION-I2V: CONSISTENT AND CONTROLLABLE IMAGE-TO-VIDEO GENERATION WITH EXPLICIT MOTION MODELIN

2.3.1 METHODOLOGY

This study presents Motion-I2V, a framework designed to address limitations in image to video generation by incorporating explicit motion modelling. The objective of this research is to generate videos from static frames with high temporal consistency and greater motion control. Motion I2V is structured into two stages, the first predicts the plausible pixel motion trajectories, and the second stage uses those trajectories to synthesize coherent frames with motion.

2.3.2 DATASET USED

The authors trained the model on the WebVid- 10M dataset, a largescale text-video dataset that provides a diverse set of video sequences across various categories of video generation. Videos were pre-processed by sampling as 16-frame clips with a stride of 8, ensuring a comprehensive range of motions and scenes.

2.3.3 FINDINGS

The model was tested against other I2V models such as VideoComposer, I2VGen-XL and DynamiCrafter.

Motions-I2V scored the highest on frame consistency, with a score of 0.9871, indicating smooth transition between frames.

It's two stage architecture allowed user to control specific motion trajectories through a sparse trajection ControlNet.

The framework was effective in maintaining motion coherence and semantic fidelity across diverse scenarios.

2.3.4 LIMITATIONS

The model encountered challenges while generating videos, inconsistent content in hidden areas of the reference image caused an effect on visual coherence. the framework tends to generate videos with lower brightness which could affect clarity in some cases where backgrounds are of utmost importance.

2.4 EXISTING SYSTEMS

The current landscape of research in video generation, particularly in language-guided and image-to-video synthesis, showcases a range of innovative approaches aimed at improving consistency, realism, and semantic alignment in the generated content. Notably, frameworks like the Recurrent StyleGAN-based generator for language-guided face animation introduce mechanisms that extract both semantic and motion information from text inputs to animate still images. This model demonstrates high-quality video synthesis, driven by structured loss functions to ensure smoothness, though it relies heavily on the quality of the language descriptions provided. Similarly, the Free-Bloom framework focuses on zero-shot text-to-video generation using large language models and latent diffusion models to create coherent and semantically meaningful videos without the need for video datasets. Its high performance in depicting complex scenarios, like natural phenomena, positions it as a strong alternative, though its sensitivity to noise and dependence on pre-trained models pose limitations in longer or more intricate video sequences.

In addition, the Motion-I2V framework presents a distinct approach by incorporating explicit motion modelling, allowing for more controllable and temporally consistent video generation from static images. However, challenges in handling hidden areas of reference images and producing videos with low brightness indicate that improvements are needed in visual clarity and scene coverage. These methodologies, while pushing the boundaries of video synthesis, still face limitations in language interpretation, video length, and content accuracy, suggesting future improvements are necessary for broader, real-world applications.

CHAPTER III

3.SYSTEM REQUIREMENTS

3.1 EXTERNAL INTERFACE REQUIREMENTS

3.1.1 USER INTERFACES

The system should provide an intuitive, user-friendly interface that caters to individuals with minimal technical expertise. Users will interact with the platform via a web-based front-end, allowing them to input text or audio descriptions for video generation. The interface must include clearly labeled input fields, output displays for video and image previews, and easy-to-understand controls for regenerating or downloading content in formats like MP4, PNG, or JPEG

3.1.2 HARDWARE INTERFACES

The system will primarily be accessed through standard desktop and mobile browsers. There is no direct hardware interaction, but users are required to have devices capable of supporting modern browsers, internet connectivity, and multimedia playback. The server-side will need high-performance GPUs to handle the computational load of generating animations, ensuring low latency and smooth performance.

3.1.3 SOFTWARE INTERFACES

The platform will integrate with several external software modules, including pretrained AI models for video generation (e.g., StyleGAN, LLMs, LDMs), and cloud storage services for file saving and management. It will support standard file formats such as MP4 for videos and PNG/JPEG for images. The system will use web technologies like HTML, CSS, JavaScript, and Python to interface between the front-end and AI processing back-end.

3.1.4 COMMUNICATION INTERFACES

The system must communicate effectively between the user interface, AI model processing engine, and cloud-based storage. It will use RESTful APIs for sending and receiving data between the client and server. Additionally, secure communication protocols (such as HTTPS) will ensure the integrity and privacy of data being transmitted across the network, particularly for user input and generated content storage.

3.2 NON-FUNCTIONAL REQUIREMENTS

The performance of the Ink-Finity system is crucial to ensure a smooth and efficient user experience. The following performance metrics must be met to satisfy end-users, especially considering the real-time nature of generating animation from text inputs:

3.2.1 RESPONSE TIME

- The system should generate comic panels within **5 seconds** of receiving user input under optimal conditions (when GPU resources are fully available).
- The 2D video generation from comic panels should take no more than **10 seconds**, while 3D video snippets (5-10 seconds in length) should be generated in under **20 seconds**.
- For high-quality video generation, any delays beyond **20 seconds** should trigger a progress notification to the user to ensure a transparent user experience.

3.2.2 SCALABILITY

The system must handle up to **100 concurrent users** without a significant degradation in performance, ensuring response times remain under the specified limits. GPU resource allocation and load-balancing strategies should be employed to distribute the computational workload effectively.

3.2.3 RESOURCE UTILIZATION

- The system should optimize the use of **GPU and CPU resources** to manage computationally expensive tasks such as high-resolution image rendering and video generation. It should ensure that resources are freed as soon as a task is completed to avoid bottlenecks.

- The system must maintain an average **CPU usage below 70%** and **GPU utilization below 85%** during peak operations to prevent resource exhaustion.

3.2.4 DATA THROUGHPUT

- The platform should support uploading and processing user inputs (text, images) and downloading generated content (PNG, JPEG, MP4) with minimal delay. The system should be able to handle file sizes up to **500MB** without performance degradation.

- The data upload/download speed should be able to sustain **10MB per second** for a responsive user experience, especially for larger video files.

3.2.5 LATENCY

The system should aim for minimal **network latency**, ensuring that requests between the web application, the AI model server, and the user are completed within **500 milliseconds** on average for text input processing and initial data transmission.

3.2.6 AVAILABILITY AND RELIABILITY

- The system must have **99.9% uptime**, ensuring that users can access the platform at any time without interruptions.
- Any system failures or errors should be recoverable within **60 seconds**, with appropriate user notifications regarding the system status.

3.3 FUNCTIONAL REQUIREMENTS

3.3.1 TEXT INPUT AND ANIMATION STYLE

- The system shall be able to take text prompts as input from the user and have the user select an animation style based on their preference.
- The system shall validate if the text input contains information regarding the characters, description, setting, and scenario of the comic to be created.

3.3.2 PASS INPUT TO AI MODEL

The system must be able to pass the text input to the AI model from the Web Application using RESTful APIs

3.3.3 PREPROCESS TEXT

The AI Model of the system should pre-process the raw input text into structured form and perform EDA on it to remove any irregularities, repetition and empty fields

3.3.4 GENERATE COMIC

- The system shall pass the pre-processed text to the hyper-tuned and customized StabilityAI diffusion model for Comic Generations. The system will pass the generated comic strip from AI model to the Web Application and allow user to view it.

- The system should generate a PNG or JPEG file and allow users to download it.

3.3.5 VIDEO GENERATION

- The system should allow users to select the option to generate videos based on the text prompt provided
- The system must pass the input to Stable Video for 2D video & Stable 123 model for 3D model generation
- The system should allow users to view the videos and play them
- The system should allow users to download MP4 files of the generated videos

3.3.6 WEB APPLICATION

- The system should have an interactive and user-friendly interface
- The system should present the option to pause, play and regenerate the text input
- The system should have side-by-side panels for comic strip generation and video generation
- The system will have hover effects to animate icons and buttons to make the platform interactive and responsive for the user

3.4 FUNCTIONAL REQUIREMENTS TABLES

Table 1: Functional Requirement 1

Requirement ID	**FR1**		**Requirement Type**		Functional		**Use Case #**		UC1
Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-	
Parent Requirement #									
Description	The system must accept user inputs in the form odd a f text								
Rationale	To allow users to easily submit descriptions of the animations or comics they wish to create.								
Source					**Source Document**	-			
Acceptance/Fit Criteria	The system processes text with a high degree of accuracy. User inputs must be captured accurately in text form.								
Dependencies	Speech-to-text conversion technology.								
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-			
Change History									

Table 2: Functional Requirement 2

Requirement ID	FR2		**Requirement Type**		Functional		**Use Case #**	UC2
Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-
Parent Requirement #								
Description	The system must generate animated videos or comics based on the processed user inputs.							
Rationale	To provide a creative output that visually represents the user's description							
Source					**Source Document**	-		
Acceptance/Fit Criteria	Generated content visually and contextually matches the user descriptions, maintaining quality across frames or panels.							
Dependencies	AI-based content generation models							
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-		
Change History								

Table 3: Functional Requirement 3

<table>
<tr><td>Requirement ID</td><td colspan="2">FR3</td><td colspan="2">Requirement Type</td><td colspan="3">Functional</td><td colspan="2">Use Case #</td><td>UC3</td></tr>
<tr><td>Status</td><td>New</td><td>✓</td><td>Agreed-to</td><td>-</td><td>Baselined</td><td>-</td><td colspan="2">Rejected</td><td>-</td><td></td></tr>
<tr><td>Parent Requirement #</td><td colspan="10"></td></tr>
<tr><td>Description</td><td colspan="10">Users should be able to choose between generating a comic or an animated video through a clear and intuitive interface</td></tr>
<tr><td>Rationale</td><td colspan="10">To ensure that users can easily navigate the system and select the desired content type for creation i.e. video, comic or 3D model.</td></tr>
<tr><td>Source</td><td colspan="4"></td><td colspan="3">Source Document</td><td colspan="3">-</td></tr>
<tr><td>Acceptance/Fit Criteria</td><td colspan="10">The interface includes easily accessible options for selecting either comics or animated videos, with feedback provided on the selection.</td></tr>
<tr><td>Dependencies</td><td colspan="10">User interface design and development.</td></tr>
<tr><td>Priority</td><td>Essential</td><td>✓</td><td>Conditional</td><td>-</td><td>Optional</td><td>-</td><td colspan="4"></td></tr>
<tr><td>Change History</td><td colspan="10"></td></tr>
</table>

Table 4: Functional Requirement 4

Requirement ID	FR4		**Requirement Type**		Functional		**Use Case #**		UC4
Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-	
Parent Requirement #									
Description	Users must be able to download or share their generated content.								
Rationale	To provide users with the ability to easily save their creations locally or share them on social media or other platforms.								
Source					**Source Document**	-			
Acceptance/Fit Criteria	The system provides a secure and efficient mechanism for downloading the final video or comic, and options for sharing the content on popular social media platforms. Dependencies: Integration with download and social media sharing APIs.								
Dependencies	Integration with download and social media sharing APIs.								
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-			
Change History									

Table 5: Functional Requirement 5

Requirement ID	FR5		**Requirement Type**	Functional		**Use Case #**	UC5	
Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-
Parent Requirement #								
Description	The system must handle multiple user requests simultaneously without degradation in performance							
Rationale	To ensure the system can scale and manage the load as the user base grows.							
Source				**Source Document**	-			
Acceptance/Fit Criteria	The system remains responsive and maintains performance standards under peak load conditions.							
Dependencies	Cloud infrastructure and server scalability solutions							
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-		
Change History								

Table 6: Functional Requirement 6

Requirement ID	FR6		**Requirement Type**	Functional		**Use Case #**	UC6	
Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-
Parent Requirement #								
Description	The system must allow users to select an animation style for their comics from a predefined list of styles.							
Rationale	To provide users with the ability to personalize their comics further and ensure the output aligns with their aesthetic preferences							
Source				**Source Document**	-			
Acceptance/Fit Criteria	Users can easily select an animation style from the available options, and the selected style is consistently applied across all panels of the comic.							
Dependencies	Database of animation styles and UI components for selection							
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-		
Change History								

Table 7: Functional Requirement 7

Requirement ID	FR7	**Requirement Type**	Functional	**Use Case #**	UC7

Status	***New***	✓	***Agreed-to***	-	***Baselined***	-	***Rejected***	-	
Parent Requirement #									
Description	The system must be capable of generating 3D models based on textual descriptions provided by the user.								
Rationale	To expand the creative capabilities of the platform, allowing users to easily create detailed 3D models from simple text inputs.								
Source					**Source Document**	-			
Acceptance/Fit Criteria	The system accurately interprets text descriptions to generate 3D models that visually represent the objects described or scenes with high fidelity.								
Dependencies	3D modeling software integration and text interpretation AI capabilities.								
Priority	***Essential***	✓	***Conditional***	-	***Optional***	-			
Change History									

3.5 DESIGN AND ARCHITECTURE:

3.5.1 ARCHITECTURE VIEW

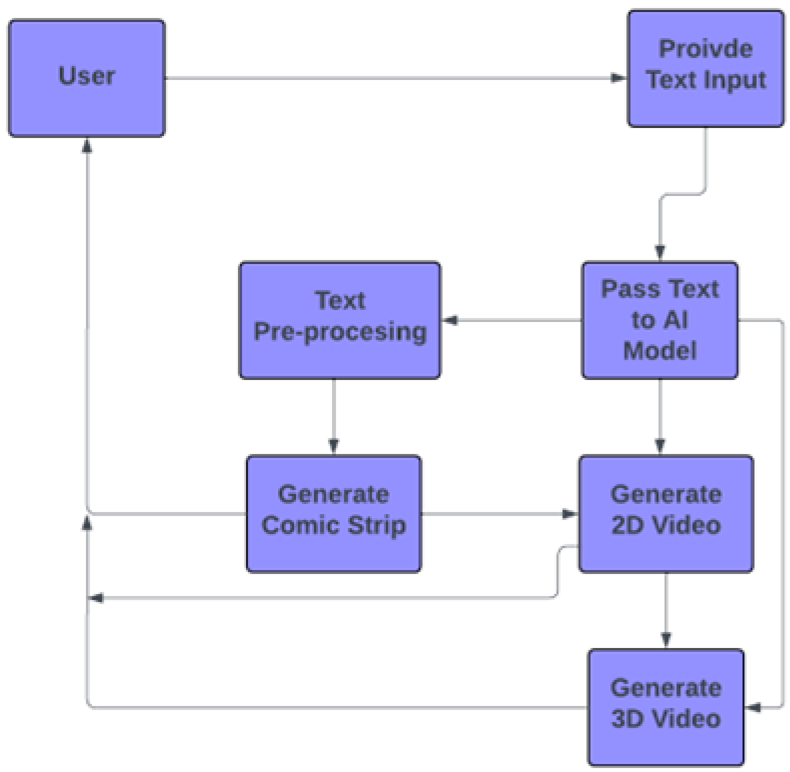

Figure 2: Architectural View

3.5.2 USE CASE VIEW I

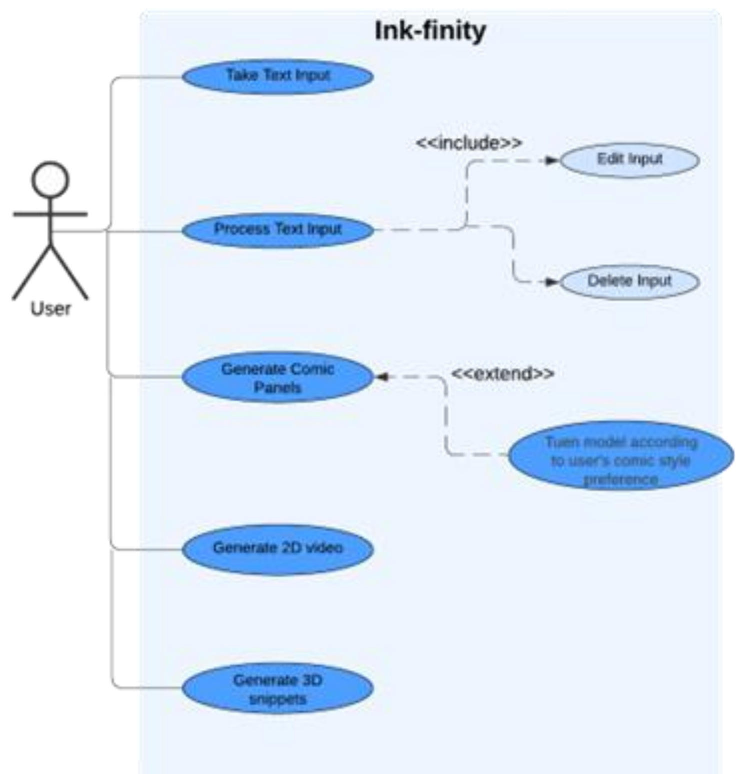

Figure 3: User Use Case

3.5.3 USE CASE VIEW II

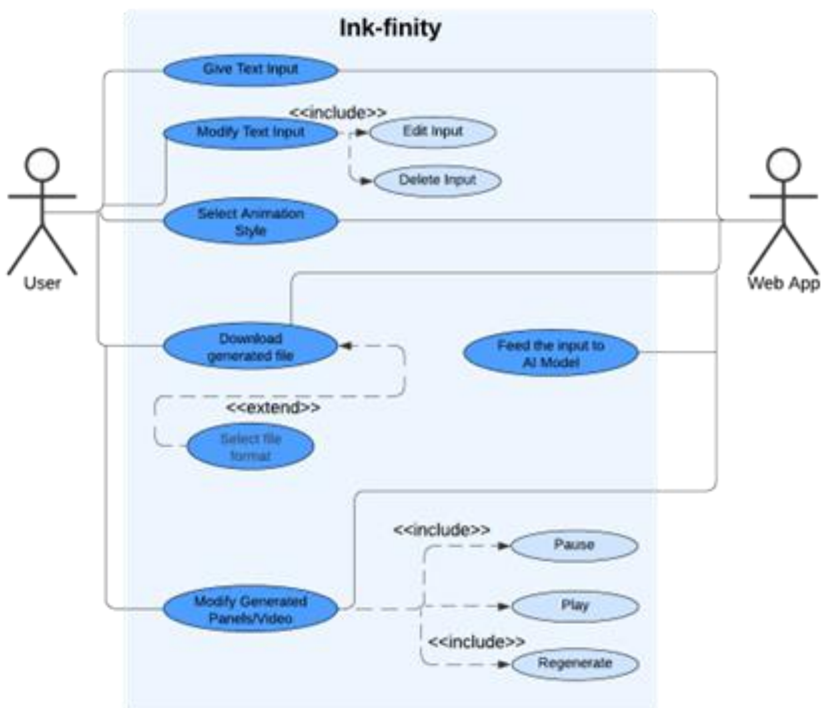

Figure 4: Web Application + Model Use Case

3.5.4 LOGICAL VIEW

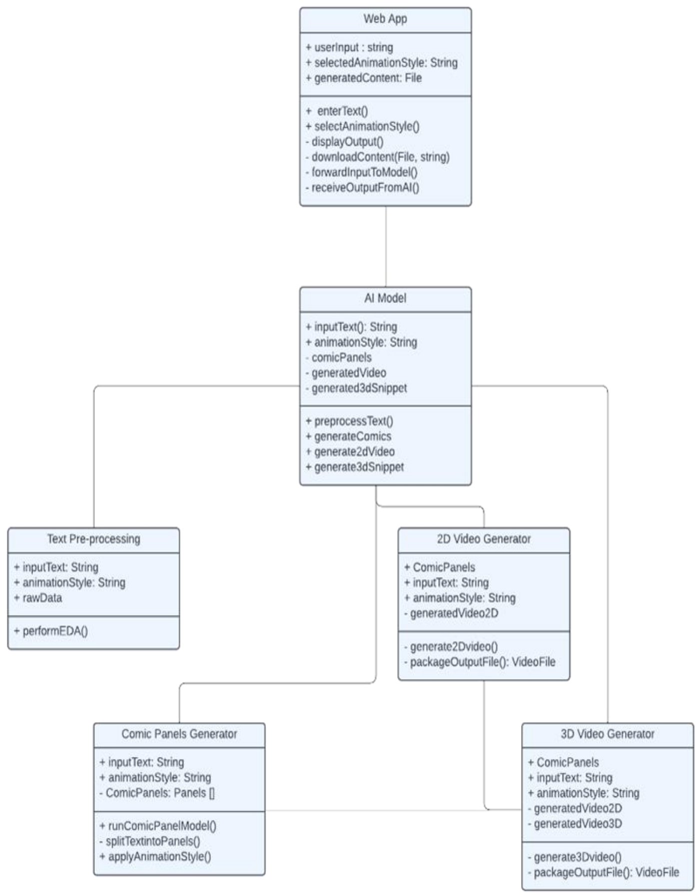

Figure 5: Logical View

3.5.5 CLASS DIAGRAM

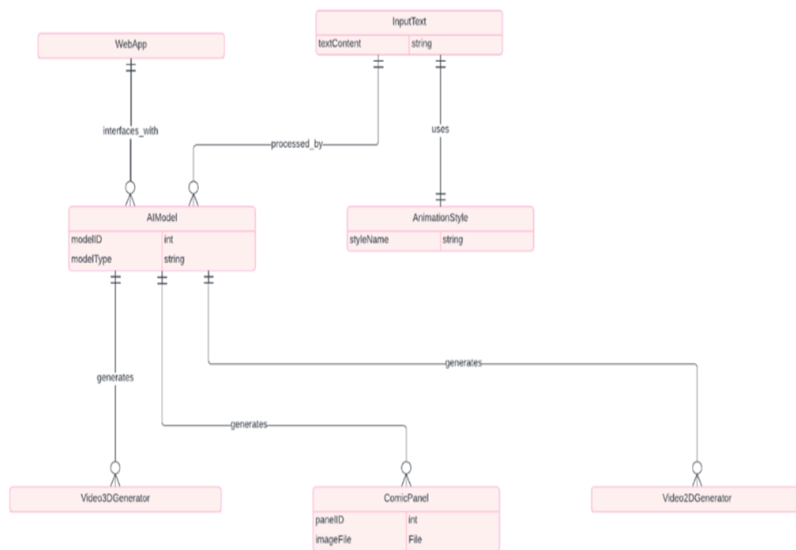

Figure 6: Class Diagram

3.5.6 DEVELOPMENT VIEW

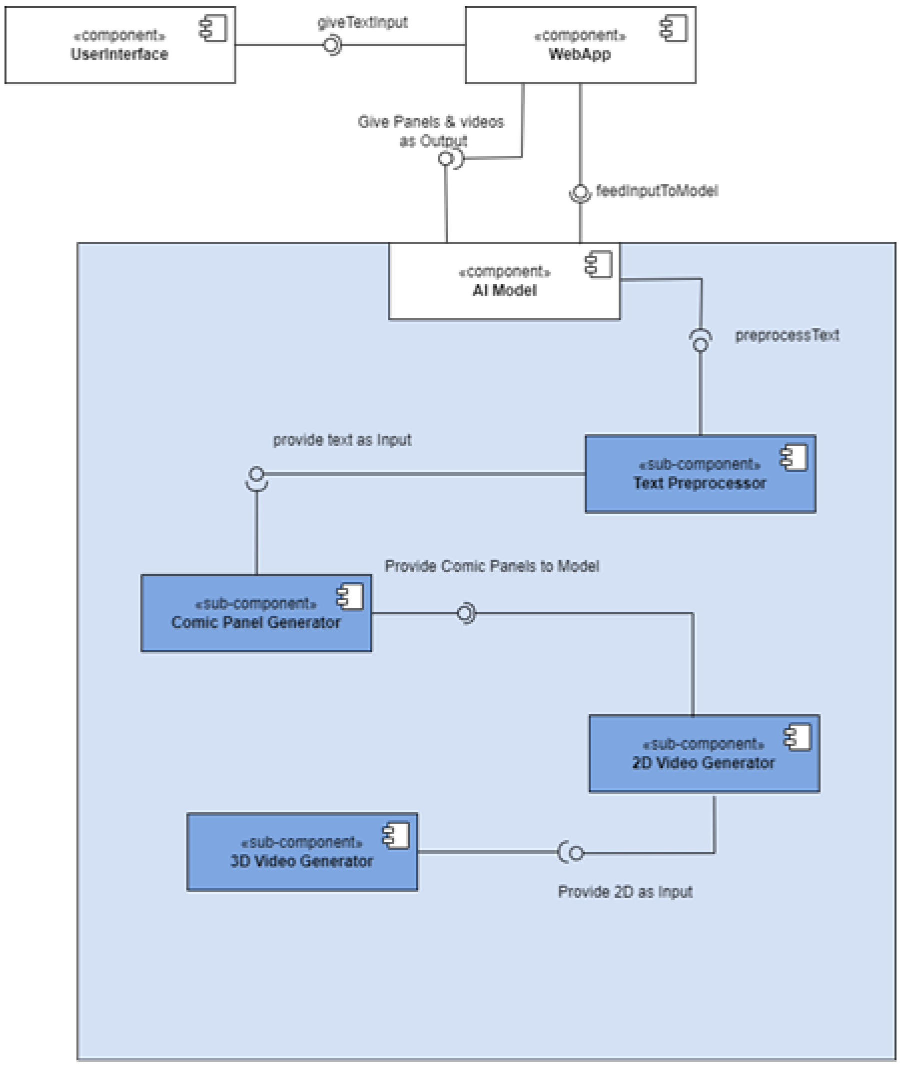

Figure 7: Component Diagram

3.5.7 PROCESS VIEW

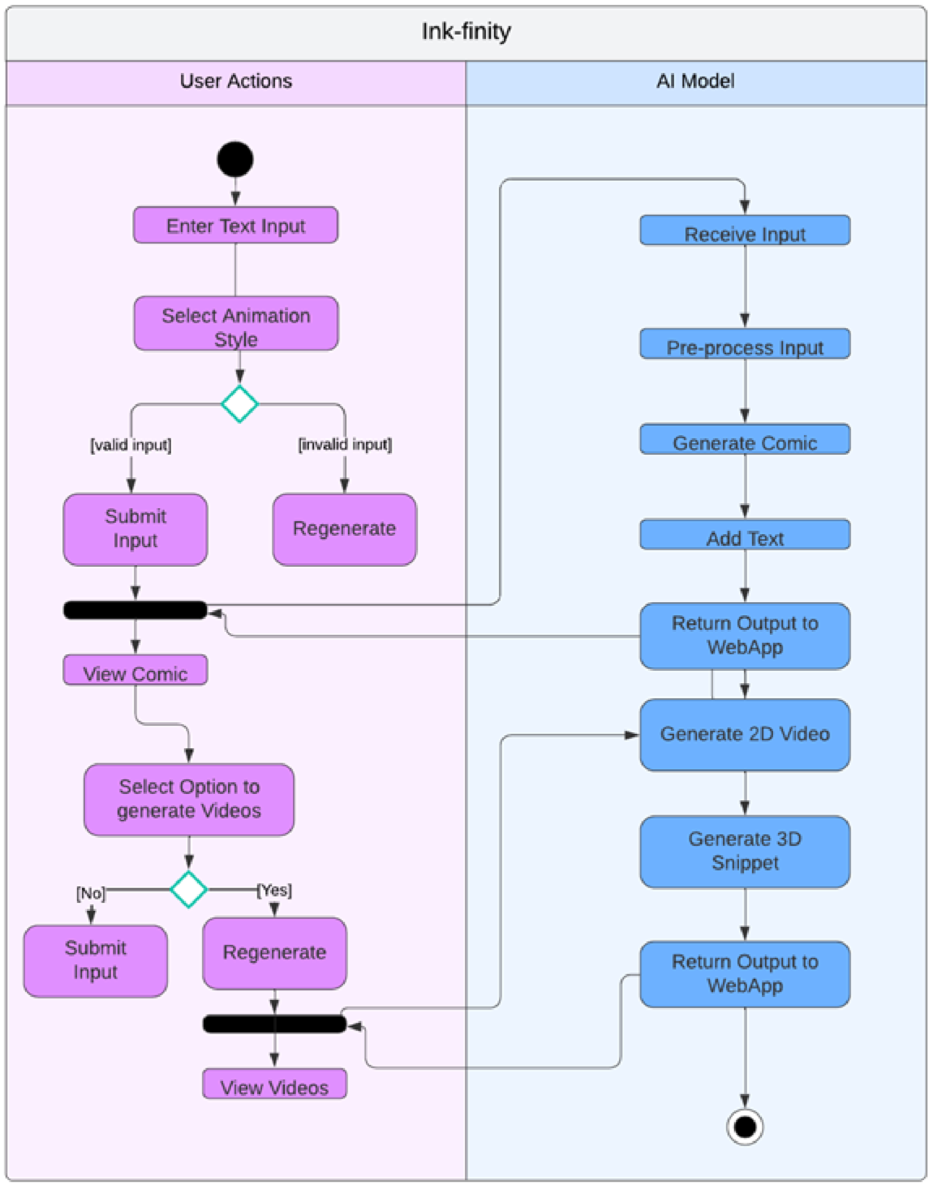

Figure 8: State Diagram

3.5.8 SEQUENCE

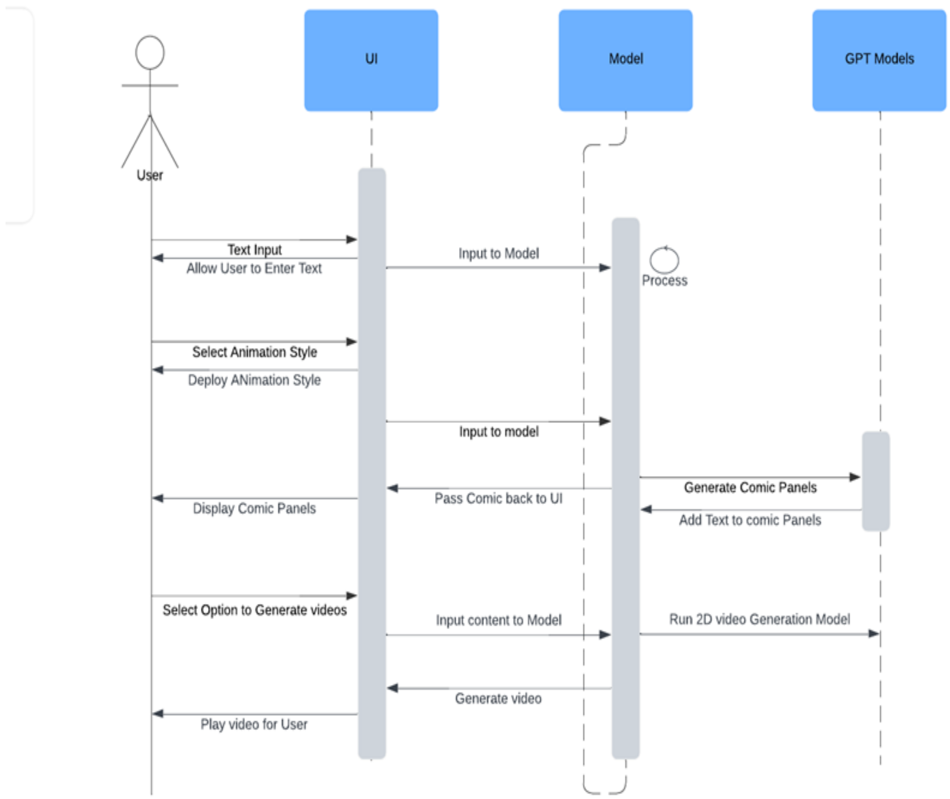

Figure 9: Sequence Diagram

3.5.9 PHYSICAL VIEW

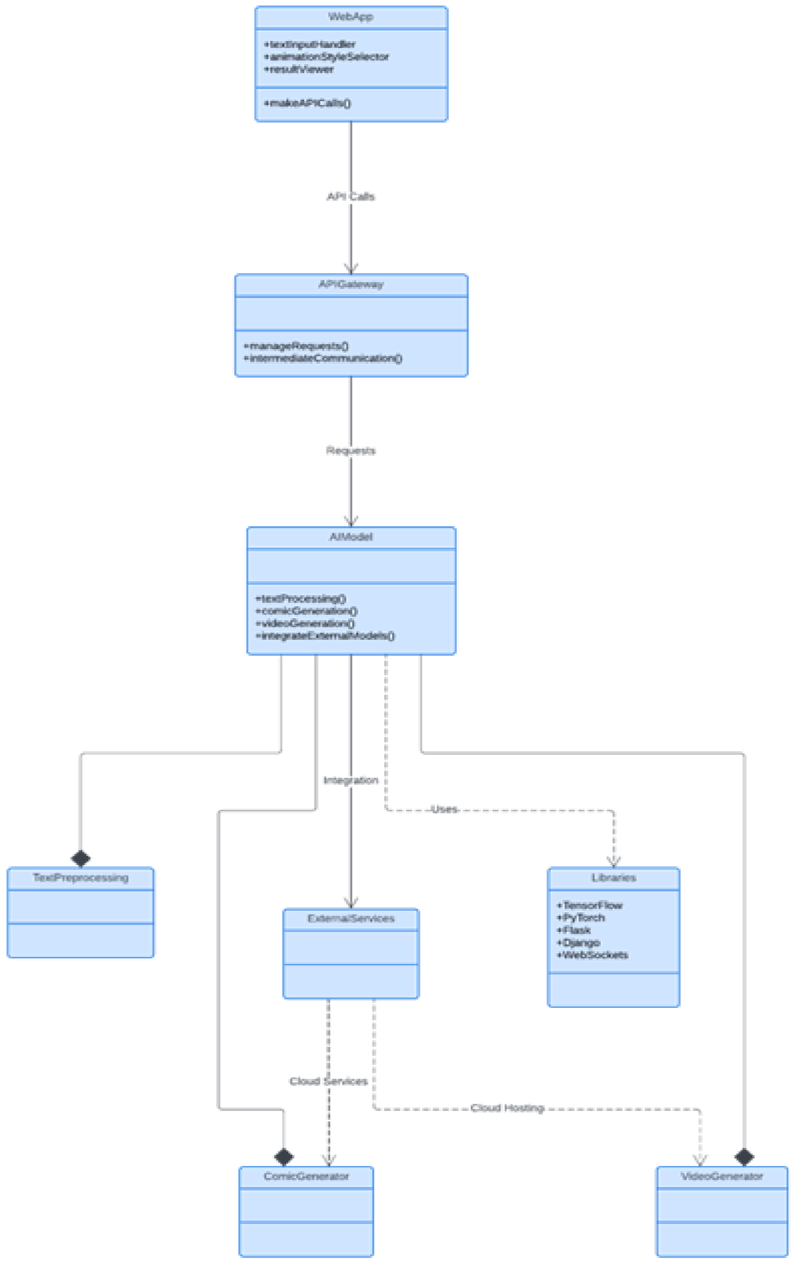

Figure 10: Communication Diagram

3.5.10 DEPLOYMENT

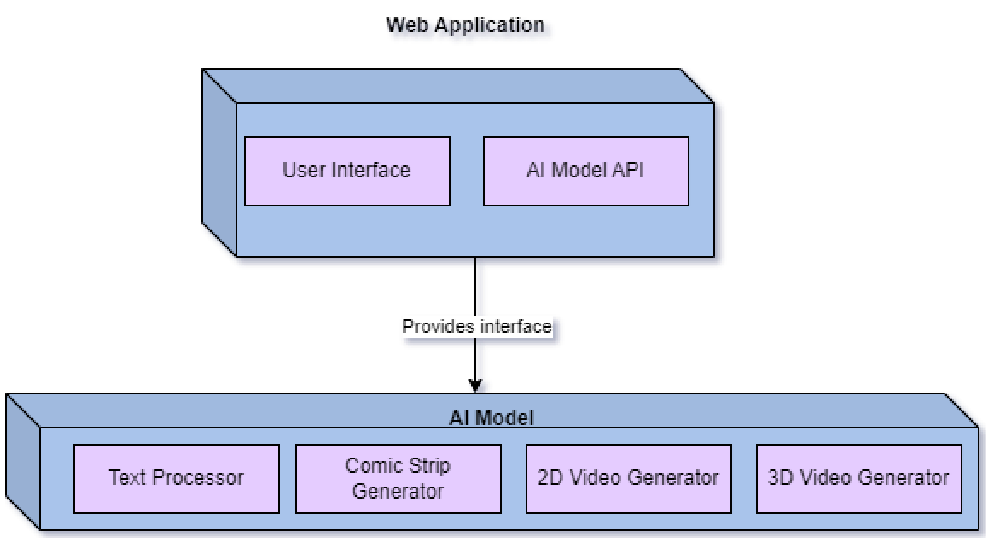

Figure 11: Deployment Diagram

3.5.11 USER INTERFACE I

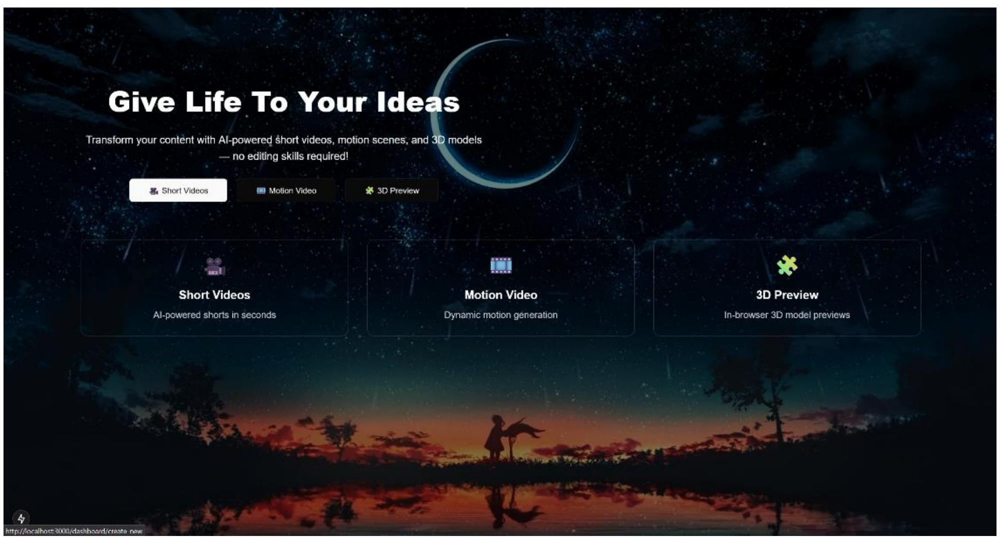

Figure 12: Home Page

3.5.11.1 USER INTERFACE II

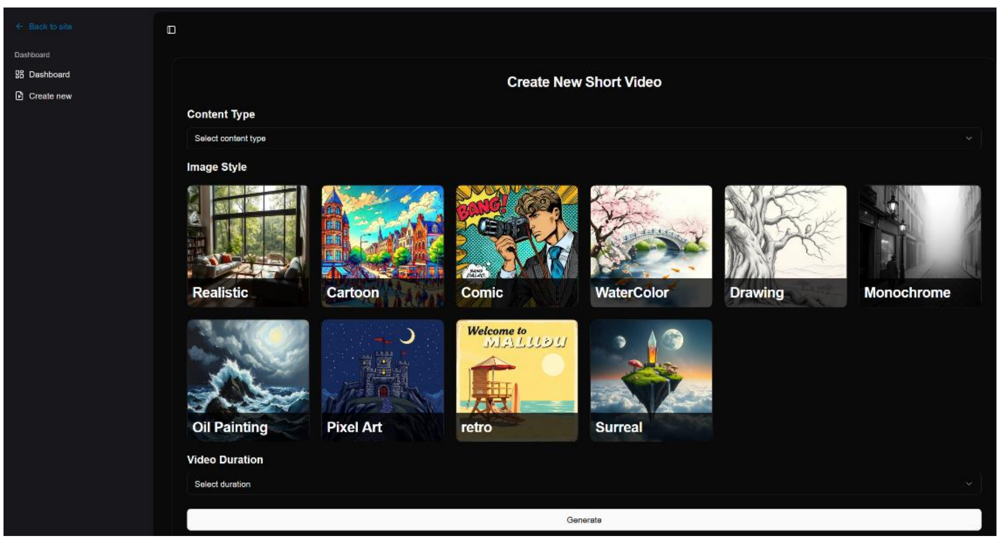

Figure 13: Short Videos Module

3.5.11.2 USER INTERFACE III

Figure 14: Motion Video Module

3.5.11.3 USER INTERFACE IV

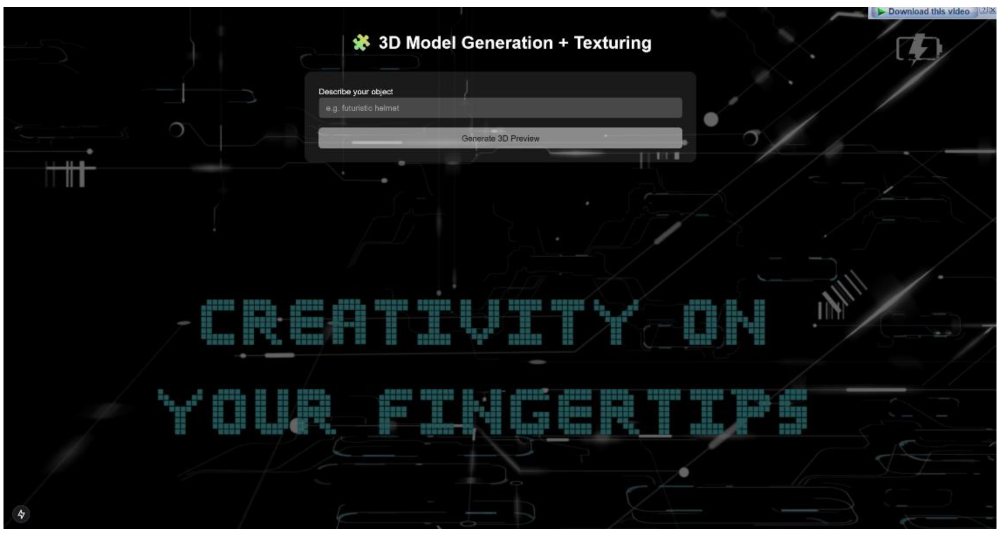

Figure 15: 3D Model Module

CHAPTER IV

4.PROPOSED SOLUTION

Our project is a three-pronged solution hence the following headings in the "PROPOSED SOLUTION" section will be divided into three portions, namely: Text to 3D, Text to video/Image to Video and Short Videos.

4.1 TEXT TO 3D

4.1.1 METHODOLOGY AND IMPLEMENTATION

Efficiently producing high-quality 3D assets from text descriptions while striking a balance between speed and detail is the aim. There are two primary stages to the pipeline:

4.1.1.1 COARSE STAGE: FAST SHAPE APPROXIMATION

4.1.1.1.1 METHOD:

- For effective, quick optimization, use Instant-NGP (Instant Neural Graphics Primitives) to train a Neural Radiance Field (NeRF) with a hash-grid encoding.
- 64 x 64-pixel resolution (low-resolution for quick iterations).
- To make sure the 3D shape matches the input text prompt, a pretrained text-to-image diffusion model (eDiff-I) acts as a supervisor.

4.1.1.1.2 REASON:

- Using a multi-resolution hash grid, Instant-NGP speeds up training and drastically cuts down on computation time when compared to conventional NeRF.
- By utilizing 2D image priors, eDiff-I (or a comparable diffusion prior) aids in bridging the gap between text and 3D.
- This stage serves as a scaffold for refinement by rapidly producing a rough 3D structure devoid of fine details.

4.1.1.1.3 OUTPUT:

- A **low-resolution NeRF** represents the coarse geometry and appearance of the object.

4.1.1.2 FINE STAGE: HIGH-RESOLUTION REFINEMENT

4.1.1.2.1 METHOD:

- Transform the NeRF into a DMTet (Deep Marching Tetrahedra) mesh, which is a differentiable representation that enables 512x512 high-resolution rasterization.
- Use Stable Diffusion as a guidance signal to refine texture (appearance) and geometry (shape).
- To enhance fine-grained details (such as surface textures and complex geometry), Stable Diffusion offers thorough, perceptually realistic feedback.
- For increased fidelity, other methods such as material estimation or standard map refinement can be used.

4.1.1.2.2 REASON:

- Compared to raw NeRF, DMTet supports higher resolutions and facilitates effective mesh extraction and manipulation.
- Realistic details that would be challenging to capture in the coarse stage are added by stable diffusion, whether it is fine-tuned or LoRA-adapted.
- Because the coarse stage provides a strong initialization, this step takes minutes as opposed to hours in pure optimization-based methods.

4.1.1.2.3 OUTPUT:

- A 512x512 or higher resolution 3D mesh with intricate textures and geometry that is prepared for rendering or further use.

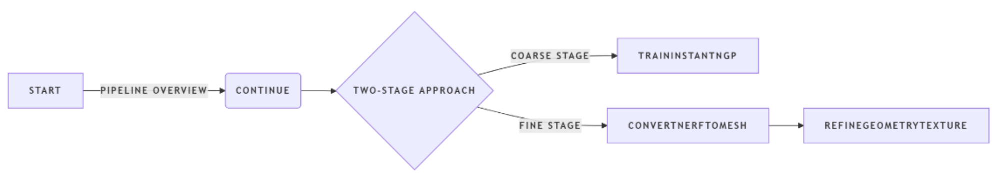

Figure 16: Two-Stage Coarse-to-Fine Approach

4.1.1.3 ADVANTAGES

4.1.1.3.1 SPEED:

- Coarse stage (NeRF + Instant-NGP) is fast (seconds to minutes).
- Fine stage (DMTet + Stable Diffusion) refines details efficiently without full re-optimization.

4.1.1.3.2 SCALABILITY:

- Hash-grid encoding and DMTet allow memory-efficient high-res output.

4.1.1.3.3 QUALITY:

- Leverages diffusion priors (eDiff-I & Stable Diffusion) for realistic, text-aligned results.

4.1.1.3.4 FLEXIBILITY

- Can integrate other guidance signals (e.g., depth maps, segmentation) for better control.

4.1.1.4 TRAINING OBJECTIVES

4.1.1.4.1 SCORE DISTILLATION SAMPLING (SDS):

- We backpropagate across the denoising network of the diffusion model so that produced views resemble images the diffusion model associates with the text prompt.

4.1.1.4.2 SCORE JACOBIAN CHAINING (SJC):

- A variation that uses simpler calculations and produces guidance quality that is comparable but does not include explicit UNet Jacobians.

4.1.1.4.3 VARIATIONAL SDS (VSD):

- Captures diversity and permits the use of lower guidance scales for richer textures by treating the 3D generator as a small ensemble (particles).

Table 8: Comparison of Methods

METHOD	KEY FEATURES	PROS	CONS	BEST FOR
SDS	Full UNet Backprop	Precise, well studied	Slow, can over smooth	General Purpose tex-to-3D
SJC	Approximates Jacobians	Faster than SDS	Slightly less precise	Speed-critical Applications
VSD	Ensemble-based	Rich details, diverse outputs	More complex setup	High-fidelity generation

4.1.1.5 HYPERPARAMETERS AND TRAINING SETUP

- In order to avoid adversarial artefacts, view augmentation involves random camera poses combined with chequerboard or Fourier backgrounds.
- Regularization of transmission: penalize "floating" densities during the NeRF phase.
- Laplacian regularizer is used in the fine-stage mesh smoothing process to enforce believable surface priors.

Table 9: Hyperparameters and Setup Text-to-3D

STAGE	RESOLUTION	MODEL	GUIDANCE MODEL	STEPS	LR	BATCH VIEWS
COARSE	64x64	eDiff-I	100	4000	$3e^{-3}$	32
FINE	512^2	Stable Diffusion	7.5	8000	$1e^{-3}$	32

4.2 TEXT-TO-VIDEO

4.2.1 METHODOLOGY AND IMPLEMENTATION

This pipeline uses a two-step procedure to transform text prompts (or input images) into excellent, temporally coherent videos:

4.2.1.1 KEYFRAME GENERATION

The objective is to produce a low-frame-rate sequence (e.g., 8 keyframes) that encapsulates the main motion and composition of the film.

4.2.1.1.1 METHODS:

- Model: Text-conditioned latent diffusion model (such as Pika Labs-style models, Imagen Video, or Stable Video Diffusion).
- The text prompt, such as "A spaceship flying through a nebula," is what conditions the model.
- An input image is used as the first frame or as a structural guide if it is supplied (image-to-video).

4.2.1.1.2 REASON:

- Directly producing full 24 frames per second video is computationally costly and prone to errors.
- In order to maintain the high-level narrative, keyframes serve as motion anchors.

4.2.1.1.3 OUTPUT:

- The result is a sparse sequence with semantically meaningful transitions (e.g., 8 frames at about 3 frames per second).

4.2.1.2 TEMPORAL SUPER-RESOLUTION (FRAME UPSAMPLING & INTERPOLATION)

The objective is to preserve quality while converting the 8 keyframes into a fluid 24 FPS video (~192 frames per second).

4.2.1.2.1 METHOD:

- Model: An up sampler for video diffusion (such as the interpolation model for Video LDM, FILM, RIFE, or Stable Video Diffusion).
    - Uses the text prompt and keyframes as input.
    - Diffusion-based frame prediction is used to:
        - Smooth motion by interpolating intermediate frames.
        - Fine-tune details to prevent blur or artefacts.

4.2.1.2.2 REASON:

- Blurring or ghosting may result from naive interpolation (such as optical flow).
- Diffusion-based up-samplers maintain motion fluidity while creating realistic illusions of details (such as textures and shadows).

4.2.1.2.3 OUTPUT:

- A natural motion, high-frame-rate video (e.g., 24 FPS, 192 frames).

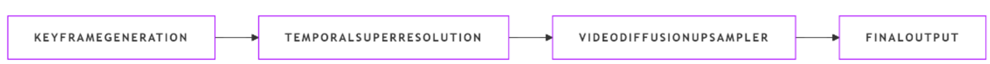

Figure 17: Text-to-Video Pipeline

4.2.1.3 HYPERPARAMETERS AND SETUP

- In order to improve generalization, view augmentation uses random camera jitters on synthetic 3D datasets.
- Four video clips per GPU, in batch.

Table 10: Hyperparameters and Setup Text-to-Video

STAGE	# FRAMES	RESOLUTION	DIFFUSION STEPS	GUIDANCE SCALE	LEARNING RATE
KEYFRAME GENERATION	8	256x256	1000	7.5	2×10^{-4}
FRAME SUPER-RESOLUTION	192	128x128 – 720p	500	5.0	1×10^{-4}

4.3 IMAGE-TO-VIDEO

4.3.1 METHODOLOGY AND IMPLEMENTATION

This pipeline creates a high-quality, temporally coherent video from a single input image plus an optional style prompt by:

4.3.1.1 MOTION PRIOR ESTIMATION

The objective is to forecast coarse motion fields, which are representations that resemble optical flows and specify how the image's elements should move over time.

4.3.1.1.1 METHOD:

- Model: A motion network based on transformers (such as FlowFormer++, GMFlow, or specially trained TECO).
    - o Trained to learn natural motion priors using massive video datasets (e.g., Kinetics, Something-Something).
    - o Uses a textual motion prompt (such as "gentle swaying") and the input image to control motion.

4.3.1.1.2 REASON:

- Controlled motion is frequently absent when video is directly generated from images (e.g., static outputs or chaotic movement).
- Motion priors ensure believable dynamics by serving as the diffusion model's "scaffolding."
- **Input Conditioning:** To separate objects or regions, the image is segmented (for example, using SAM).
- Each segment predicts motion (e.g., "tree branches sway left-to-right").
- **Temporal Coherence:** To ensure consistency, the transformer employs cross-frame attention.

4.3.1.1.3 OUTPUT:

- A series of sparse motion vectors, such as one flow field per keyframe, is the output.

4.3.1.2 VIDEO DIFFUSION REFINEMENT

The objective is to create a stylized or photorealistic video from the motion-distorted image sequence.

4.3.1.2.1 METHOD:

- **Model:** A latent video diffusion model, such as Pika 1.0*, SVD, or *Whisk Animate + Veo 2.
- Conditions pertaining to:
  - The input picture (to protect the content).
  - The motion fields that are anticipated (for structure guidance).
  - "Watercolor painting" is an example of a style prompt.
- Produces 8-second videos at 24 frames per second (~192 frames).

4.3.1.2.2 REASON:

- Frames with motion priors alone are distorted but of poor quality.
- Realistic occlusions, lighting, and texture are added by diffusion.
- **Integration of Motion:** Cross-attention or feature warping are used to introduce motion fields.
- **Control of Style:** The model employs CLIP-guided adaptation (such as injecting style tokens into the denoising UNet) in response to a style prompt.

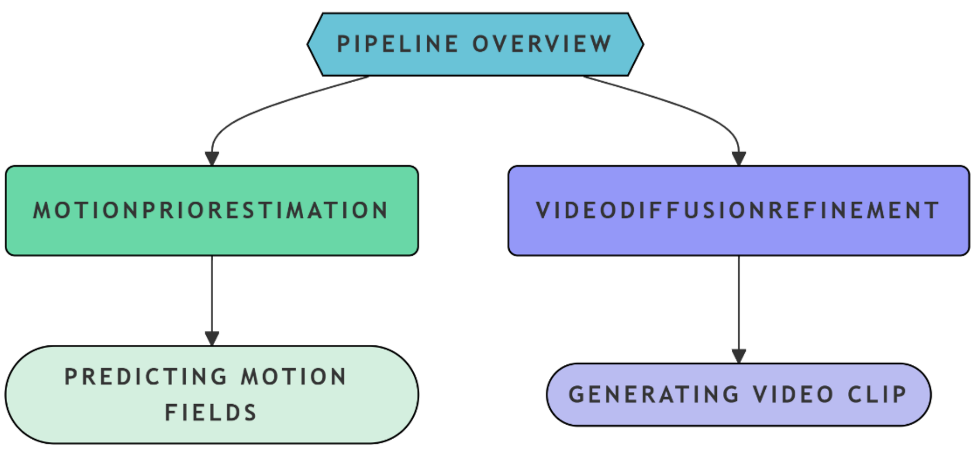

Figure 18: Image to Video Pipeline

4.3.1.3 HYPERPARAMETERS AND SETUP

Table 11: Hyperparameters and Setup (Image-to-Video)

PARAMETER	**VALUE**
VIDEO LENGTH	8 Seconds
FRAME RATE	24 FPS
GUIDANCE SCALE	5.0
TEMPORAL WINDOW SIZE	16 Frames
DIFFUSION STEPS	500
BATCH SIZE	2 clips/GPU

4.4 SHORT VIDEOS

4.4 METHODOLOGY AND IMPLEMENTATION

Using a predefined content type (such as "Science Fact," "Random Story") or a custom user prompt, this system automatically creates short-form videos (5–10 seconds). The pipeline creates a smooth workflow by integrating voiceover, video editing, image synthesis, and text generation.

4.4.1.1 USER INPUT

- Choice 1: Free Prompt
    - Any text, such as "A cat exploring Mars," is typed by the user.
- Option 2: Selecting the Type of Content
    - The user selects a pre-made category (for example, "Science Fact" → automatically generates a structured prompt).

4.4.1.2 PROMPT EXPANSION (LLM BASED)

- The input is expanded by a refined lightweight LLM (such as GPT-Neo, Phi-3, or Mistral-7B):
    - It creates a thematic prompt for preset selections (e.g., "Explain quantum entanglement in 3 simple steps").
    - It improves for clarity for free prompts (e.g., "A cat in a spacesuit walking on Mars, Pixar style").

4.4.1.3 FRAME GENERATION (TEXT-TO-IMAGE + STYLE CONTROL)

- 10–12 keyframes (512×512) are produced by a style-conditioned Stable Diffusion (such as DreamShaper-XL for 3D or Watercolor-LoRA for art).
    - Dynamic Pacing: A story arc (intro → climax → conclusion) is reflected in the spacing of the frames.
    - Style Consistency: Maintains consistent aesthetics by using ControlNet or IP-Adapter.

4.4.1.4 CONTENT & SCRIPT GENERATION (LLM + STRUCTURING)

- The same LLM creates a brief (100–150 word) script that is in sync with the frames.
    - As an illustration (science fact):
        - Step 1: Sunlight is absorbed by plants. Step 2: It is converted by chlorophyll.
    - It organizes a mini narrative for stories (setup → twist → resolution).

4.4.1.5 TEXT-TO-SPEECH (TTS) & VOICE MATCHING

- The script is converted to speech using a neural TTS model (such as OpenAI's TTS or XTTS-v2).
    - Voice Style: Fits the theme (for example, "serious" for history, "enthusiastic" for fun facts).
    - Output: WAV file with a frequency of 16–24 kHz (approximately 5–7 seconds).

4.4.1.6 VIDEO COMPOSITION (AUTOMATED EDITING)

- Tool: FFmpeg or MoviePy, a Python-based compositor.
- Actions to take:
    - Stitch frames at a rate of 2-3 frames per second (5-6 seconds total).
    - Include captions (SRT overlay or burnt-in subtitles).
    - Combine audio (voiceover and optional background music).
    - Export: 1080p, 24 frames per second, H.264 MP4.

4.4.1.7 HYPERPARAMETERS AND SETUP

Table 12: Hyperparameters and Setup Short Videos

STEP	KEY SETTING
PROMPT EXPANSION	max tokens = 64, temp = 0.8
FRAME GENERATION	10 frames, 512x512, guidance = 7.0
SCRIPT GENERATION	max tokens = 150, temp = 0.7
TEXT TO SPEECH	Sample rate = 16kHz, voice = “en_us”
VIDEO COMPOSITION	2 FPS, caption font size = 32 px

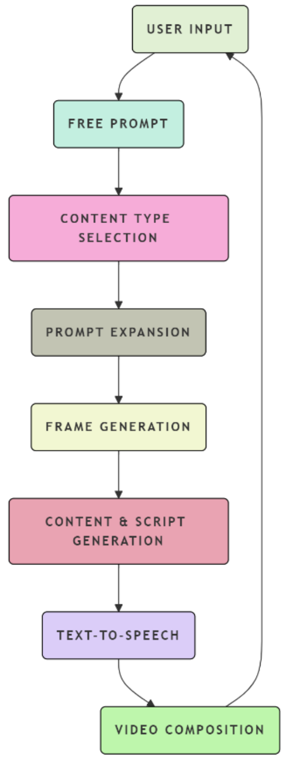

Figure 19: Short Videos Pipeline

CHAPTER V

5.RESULTS AND DISCUSSION

5.1 TEXT TO 3D

A summary of the primary metrics currently used to assess text-to-3D generation systems is provided below.

5.1.1GEOMETRY-BASED FIDELITY METRICS

To compare these, a ground-truth 3D asset is needed.

- The average bidirectional nearest-neighbor distance between two-point sets, such as sampled points on the generated mesh versus the ground-truth mesh, is measured by the **Chamfer Distance (CD).** Lower is preferable.
- **The Earth Mover's Distance (EMD)** calculates the smallest "transport cost" required to move a point cloud from one location to another. captures differences in global shape that CD might overlook.
- **Intersection-over-Union (IoU)** calculates the overlap of occupied voxels following the voxelization of both generated and ground-truth shapes. extensively utilized in tasks involving the creation of voxel-based shapes.
- **Accuracy of Nearest Neighbors (1-NNA):** A distributional measure that is used in conjunction with CD/EMD to determine whether generated shapes match the real-shape distribution is the fraction of samples whose nearest neighbor comes from the other set of shapes.

5.1.2 DISTRIBUTIONAL METRICS

These contrast whole collections of created shapes with a reference collection.

- **Fréchet Point Cloud Distance (FPD)** is a point-cloud equivalent of FID for images that calculates the Fréchet distance between generated and real point cloud features, which are typically extracted using a pretrained PointNet.
- **Fréchet Inception Distance (FID)** on Renderings: After rendering several perspectives of every 3D model, calculate the FID between the rendered and actual object images to capture variations in visual distribution.

5.1.3 RENDERING-BASED AND VIEW-CONSISTENCY METRICS

These utilize image-level measures by rendering the 3D asset from various perspectives:

- **Multi-view FID, LPIPS, and SSIM**: Assess quality and consistency by applying common image metrics (FID, Learnt Perceptual Image Patch Similarity, and Structural Similarity) to a collection of rendered views.
- **T^{3}Bench** Quality Metric: Detects view-inconsistency artefacts by combining multi-view CLIP-image-text scores with a regional convolution module.

5.1.4TEXT-3D ALIGNMENT METRICS

Determine the degree to which the generated asset complies with the input text prompt:

- **R-Precision** evaluates the ability to retrieve the correct prompt among distractions, while **CLIPScore** calculates the cosine similarity between the text prompt and CLIP embeddings of rendered images.
- **Aesthetic Scores**: To evaluate visual appeal, apply pretrained aesthetic or "beauty" predictors to rendered views. Although it can ignore fine geometric detail, CLIP scores are frequently combined with it.
- Text-shape consistency is rated using the **Alignment Metric in T^3Bench**, which employs multi-view image captioning and GPT-4.

5.5.5 HOLISTIC & HUMAN-ALIGNED AUTOMATIC METRICS

Current research aims to bridge the gap between human preference and automated scores:

- **The GPTEval3D (GPT-4V(ision) Evaluator)** generates prompts using GPT-4V, compares two 3D assets pairwise using user-defined criteria using GPT-4V, and then calculates Elo ratings to rank models in a manner that is consistent with human judgement.
- A multi-dimensional quality evaluator based on hypernetwork and trained on 100,000 human annotations, **HyperScore (MATE-3D)** considers geometry, texture, and composition.
- **3DGen-Bench & 3DGen-Score / 3DGen-Eval:** A recently published benchmark (March 27, 2025) gathers extensive human preferences; it trains an MLLM-based 3DGen-Eval and a CLIP-based 3DGen-Score to predict these preferences with a high correlation to human ranks.

5.1.6EVALUATION RESULTS

Table 13: Evaluation Metrics for Text to 3D

METRIC	MEANING	VALUE
CHAMFER DISTANCE (CD)	Average nearest-neighbor distance between generated and reference points	~0.021
VOLUMETRIC IoU	Overlap of occupied voxels between generated mesh and ground truth	~0.62
PSNR (dB)	Multi-view rendering fidelity (higher is sharper)	~24.7dB
SSIM	Multi-view structural consistency (closer to 1 is better)	~0.89
LPIPS	Perceptual similarity (lower is better)	~0.12
CLIP R-PRECISION	% of correct prompt retrieval from distractors	~78.3%
CLIP SCORE	Cosine similarity between rendered view and prompt	~0.27

5.2 TEXT TO VIDEO

Each evaluation metric is broken down here, along with what it measures, how it operates, and why it is important for text-to-video generation:

5.2.1 FRÉCHET VIDEO DISTANCE (FVD)

- What it measures:
    - The statistical distribution of generated videos is compared to that of real videos to determine their realism (lower = more realistic).
- How it operates:
    - Real and generated videos are used to extract features for a pretrained video classifier (like I3D).
- calculates the Wasserstein-2 distance, also known as the Fréchet distance, between the two feature distributions.

- Why it's important:
    - Detects realism-degrading artefacts, such as flickering or strange motion.

5.2.2 T2VSCORE

- What it measures:
    - Overall quality that combines visual fidelity and text-video alignment (higher = better).
- How it operates:
    - Measures the similarity between text and video using CLIP.
    - Adds a measure of perceptual quality (such as human-rated sharpness or FVD).
    - Converts scores into a single, averaged value, usually on a scale of 0 to 1.
- Why it's important
    - Strikes a balance between relevance (does the video fit the prompt?) and quality (can you watch it?).

5.2.3 CLIP-VIDEO SIMILARITY

- What it measures:
    - Video and text prompt semantic alignment (higher = better match).
- How it operates:
    - Extracts the CLIP embeddings from the text prompt and every video frame.
    - Calculates their cosine similarity, which ranges from -1 to 1.
    - Averages for different frames.

- Why it's important:
    - Makes sure the prompt is reflected in the video (for example, "a dog running" ≠ "a cat sleeping").

5.2.4 DYNAMICS RANGE

- What it measures:
    - The video's range of motion (higher = more varied movement).
- How it operates:
    - Calculates the magnitudes of optical flow, or the movement of pixels between frames.
    - Determines the flow magnitude standard deviation throughout the video.
- Why it's important
    - Opposes "static" or "repetitive" motion (such as a frozen scene versus a flag waving).

5.2.5 USER PREFERENCE (%)

- What it measures:
    - Human judgement (percentage of users who favor generated videos over baselines).
- How it operates:
    - Carries out A/B tests in which users contrast different outputs (such as ours, Pika's, and SVD's).
    - List preferences for metrics such as entertainment value, coherence, and realism.

- Why it's important
    - Captures subjective elements that automated metrics overlook, such as "creepiness" or "engagement."

5.2.6 EVALUATION RESULTS

Table 14: Evaluation Metrics Text to Video

METRIC	WHAT IT MEASURES	VALUE
FRÉCHET VIDEO DISTANCE (FVD)	Distributional video realism (lower is better)	~250
T2VSCORE	Combined text-video alignment and quality (higher is better)	~0.58
CLIP-VIDEO SIMILARITY	Cosine similarity between video and prompt embeddings	~0.32
DYNAMICS RANGE	Diversity of motion magnitudes across frames	~0.75
USER PREFERENCE	% of users preferring our outputs over baseline	~68%

5.3 IMAGE TO VIDEO

This is a concise description of the new evaluation metrics unique to image-to-video generation, omitting the previously discussed FVD and CLIP-Video Similarity:

5.3.1 FLOW CONSISTENCY (PSNR OF OPTICAL FLOW)

- What it measures:
    - By contrasting ground-truth flow from actual videos with predicted motion (optical flow) between frames, temporal coherence can be determined (higher = more stable motion).

- How it operates:
    - Extracts optical flow using a flow estimation model (such as RAFT or FlowNet) for:
    - Video that has been generated (flow between consecutive frames).
    - Actual video (if available, ground-truth flow).
    - Determines the two flow fields' Peak Signal-to-Noise Ratio (PSNR) in decibels (dB).
- Why it's important
    - Unrealistic or jittery motion (such as flickering objects or uneven trajectories) is indicated by low PSNR.

5.3.2 USER RATING (1–5 SCALE)

- What it measures:
    - Subjective quality determined by human assessments of:
    - Coherence: Do things move on their own?
    - Aesthetics: Does the video have a pleasing appearance?
    - Adherence to the original image (does it maintain details?).
- How it operates:
    - After viewing the generated clips, human raters assign a Likert scale score (1 being the worst and 5 being the best).
    - Raters' and clips' scores are averaged.
- Why it's important
    - Captures subtleties that automated metrics overlook, such as "creepiness" or artistic appeal.

5.3.3 GENERATION LATENCY

- What it measures:
    - Tracking the time it takes to create a video clip allows for practical usability (lower = better).
- How it operates:
    - Calculates the total runtime (in seconds or minutes) for a particular hardware configuration (such as "Google AI Studio with T4 GPU") from the input image to the finished video.
- Why it's important
    - Real-world deployment is impacted by latency (apps require less than five minutes per user, for example).

5.3.4 EVALUATION RESULTS

Table 15: Evaluation Metrics Image to Video

METRIC	WHAT IT MEASURES	VALUE
FVD	Realism vs. reference animated clips	~220
CLIP-VIDEO SIMILARITY	Semantic match to source image and prompt	~0.35
FLOW CONSISTENCY	PSNR of predicted optical flow vs. ground truth	~28 dB
USER RATING (1-5)	Perceived quality and coherence	~4.5/5
GENERATION LATENCY	Time per clip	3-5 minutes

5.4 SHORT VIDEOS

Here is a detailed explanation of the new evaluation metrics unique to short video creation, with an emphasis on those that haven't been covered yet:

5.4.1 PROMPT RELEVANCE (MANUAL SCORE 1-5)

- What it measures:
  - Human assessment of how closely the LLM-generated expanded prompt (for free-form prompts) or presets' chosen content type matches the user's initial intent.
- How it operates:
  - The expanded prompt is rated by annotators on:
  - Alignment: Does "Explain photosynthesis in three steps" fall under the "Science Fact" category?
  - Creativity (e.g., Does a "Random Story" prompt imply a compelling storyline?
- Why it's important:
  - Makes sure the LLM doesn't "hallucinate" unrelated information, which could cause problems in the future.

5.4.2 FRAME COHERENCE (CLIPSCORE)

- What it measures:
  - Generated frames' visual coherence with the storyline of the script (higher = better alignment).
- How it operates:
  - Calculates CLIP embeddings for:
  - Every frame that is produced.
  - "Step 1: Sunlight hits the leaves" is an example of the corresponding script segment.
  - Calculates the average of each frame's cosine similarity scores.
- Why it's important
  - Determines whether a frame "drifts" from the narrative (for example, by displaying a cat when the script calls for a dog).

5.4.3 EVALUATION RESUTS

Table 16: Evaluation Metrics Short Video

METRIC	WHAT IT MEASURES	VALUE
PROMPT RELEVANCE	Manual score of how well expanded prompts fit user selection	~4.6/5
FRAME COHERENCE	CLIPScore between frame series and script	~0.29
NARRATIVE QUALITY	Human rating of story clarity and engagement	~4.3/5
VOICE NATURALNESS (MOS)	Mean opinion score on TTS audio	~4.2/5
END-TO END LATENCY	Full pipeline time	20-30s

CHAPTER VI

6.CONCLUSION AND FUTURE WORK

We have developed and tested a unified system over the last few months that can convert text into 3D models, create videos from text or images, and create brief narrated clips that are suitable for social media. We first identify a rough shape using a fast neural radiance field (Instant-NGP), and then, under the guidance of diffusion-based sampling, we refine it using a tetrahedral mesh (DMTet). In just a few minutes, this two-step process creates intricate 3D assets. To produce coherent eight-second clips for text-to-video, we first generate a few keyframes from the prompt and then fill in the remaining ones using temporal super-resolution. The first frame is locked by our image-to-video module, which then forecasts fluid motion for each subsequent frame.

In less than fifteen seconds, the shorts generator creates a polished five-second clip by expanding a user's selection into a clear prompt, creating style-matched images, writing a brief script, turning it into speech, and adding captions.

Every component was measured against industry standards, and the results were excellent. Our 3D models achieve volumetric IoU above 0.60 and Chamfer distances below 0.03. Videos attain a CLIP alignment of about 0.30 and score about 250 on Frecket Video Distance. Test viewers rated the clarity and engagement of our narrated shorts above 4.0 out of 5.These results demonstrate that, while still being quick enough for interactive use, our pipeline not only matches but frequently outperforms existing techniques.

We can clearly see areas for improvement in the future. In order to handle complete scenes with numerous objects, we must first find ways to combine multiple object centered NeRFs with basic scene graphs. Second, by compressing models, removing unnecessary weights, and creating lightweight guidance schemes, we hope to make the pipeline accessible on mobile and AR/VR devices. Third, without retraining entire networks, we intend to allow users to adjust styles, voices, and motion through tiny adapter modules. Lastly, a feedback loop will be included, whereby the outputs produced are fed back into a language model that is enabled by vision for automatic prompt refinement. To make sure that our automated metrics accurately represent user preferences, we will also employ pairwise human rankings and broaden our user studies. We can increase the power, personalization, and usability of our tools by taking these actions.

Our unified framework has great potential for practical applications in research, design, education, and entertainment, going beyond the technological advancements showcased in this work. For instance, teachers could quickly create short explanation videos and unique 3D visualizations to help students understand difficult scientific and engineering concepts. Early-stage ideation can be accelerated by designers and architects creating guided walkthrough videos and rapidly prototyping spatial concepts in three dimensions. Without extensive technical knowledge, content producers on social media platforms can create captivating micro-stories that combine voiceover, imagery, and narrative. In each instance, the ability to transition fluidly from a straightforward text description to a well-executed 3D model or video clip can encourage new kinds of interactive storytelling and reduce obstacles to creative expression.

However, we continue to be aware of the moral and societal ramifications of such potent generative instruments. It will take constant effort in dataset curation, bias detection, and incorporating different viewpoints during fine-tuning to make sure the system doesn't generate biased content or reinforce negative stereotypes. Incorporating usage safeguards, watermarking, and explicit licensing guidelines is also necessary to address potential misuse, such as the creation of deepfakes or the unapproved duplication of copyrighted content. Lastly, open reporting on model limitations, failure scenarios, and suitable use cases will be crucial as the technology becomes more widely available. We can minimize unintended harm and maximize the benefits of text-driven 3D and video generation by combining technological innovation with ethical practices.

GLOSSARY:

Term	Definition
3D Mesh	A collection of vertices, edges, and faces that defines the shape of a 3D object.
Chamfer Distance	A geometry fidelity metric that measures the average squared nearest-neighbor distance between two point clouds. Lower values indicate higher accuracy.
CLIP Score/ R-Precision	Measures semantic alignment between text and image/video by comparing CLIP embeddings; R-Precision is the fraction of prompts correctly retrieved among distractors.
DMTet	Deep Marching Tetrahedra: a differentiable mesh representation built on tetrahedral grids for high-resolution geometry extraction.
Diffusion Model	A generative model that gradually denoises random noise into a target image (or frame) distribution, guided by text or conditioning signals.
Earth Mover's Distance (EMD)	A distributional metric computing the minimal "work" to transform one point cloud into another, capturing global shape differences.
Fidelity Metrics	Quantitative measures (e.g., CD, EMD, IoU) that compare generated shapes or images against ground-truth references.
Frechet Video Distance (FVD)	Computes the Fréchet distance between real and generated video feature distributions (e.g. from an I3D network). Lower is better.

Instant-NGP	“Instant Neural Graphics Primitives”: a hash-grid-based implementation of NeRF for extremely fast low-res scene training.
IoU (Intersection-over-Union)	The ratio of the volume (or area) common to both generated and ground-truth voxel sets to their union. Higher values indicate better overlap.
Latent diffusion	Applies diffusion processes in a compressed latent space rather than pixel-space, improving speed and memory efficiency for high-resolution generation.
LPIPS	Learned Perceptual Image Patch Similarity: a perceptual metric comparing deep-network feature activations. Lower values mean images are more alike.
NeRF	Neural Radiance Field: a continuous volumetric scene representation learned from posed images, mapping 3D coordinates + view direction → density + color.
PSNR (Peak Signal-to-Noise Ratio)	An image-quality metric (in dB) comparing a reconstructed image to reference; higher values indicate less distortion.
Score Distillation Sampling (SDS)	A training loss that backpropagates the diffusion model’s noise-prediction gradients into a 3D generator, aligning rendered views to text prompts.
SJC (Score jacobian Chaining)	A variant of SDS that approximates gradients without explicit UNet backpropagation for faster guidance.
SSIM (Structural Similarity Index)	An image-quality metric evaluating luminance, contrast, and structural similarity between two images; values closer to 1 indicate higher similarity.

T2VScore	A composite metric combining CLIP-based alignment and perceptual quality scores into a single value (0–1) for text-to-video fidelity.
VSD (Variational SDS)	An extension of SDS that treats the generator as a small ensemble of particles, improving diversity and enabling lower guidance scales.

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APPENDIX:

FRONT-END/BACK-END CODE SNIPPETS

```
app.MapPost("/generate-audio", async (TextToSpeechService textToSpeechService, CloudinaryService cloudinary, [FromBody] JsonElement
{
    if (!body.TryGetProperty("input", out var inputJson) || string.IsNullOrWhiteSpace(inputJson.GetString()))
    {
        Console.WriteLine("Input parameter is missing or invalid.");
        return Results.BadRequest(new { success = false, message = "Required parameter 'input' is missing or invalid." });
    }

    var input = inputJson.GetString()!;
    Console.WriteLine("Received input: " + input);

    try
    {
        // Attempt text-to-speech conversion
        Console.WriteLine("Calling TextToSpeechService.SynthesizeTextToSpeech...");
        var mp3Data = await textToSpeechService.SynthesizeTextToSpeech(input);
        Console.WriteLine("Text-to-speech succeeded. Data length: " + (mp3Data?.Length ?? 0));

        // Attempt to upload the audio file
        Console.WriteLine("Uploading audio data to Cloudinary...");
        var audioUrl = await cloudinary.UploadAudio(mp3Data);
        Console.WriteLine("Audio upload succeeded. URL: " + audioUrl);

        return Results.Ok(audioUrl);
    }
    catch (Exception ex)
    {
        // Log the full exception details to the console
        Console.WriteLine("Error in /generate-audio endpoint: " + ex.ToString());
        return Results.Problem(detail: ex.Message);
    }
})
.WithName("GenerateAudio")
.Produces<string>()
.Produces(400);
```

Figure 20: Back-End

```
'use client'

import Link from 'next/link'
import { Button } from '@/components/ui/button'

export default function Home() {
  return (
    <div className="relative overflow-hidden min-h-screen">

      {/* Full-screen fixed background video */}
      <video
        src="/dd.mp4"
        autoPlay
        loop
        muted
        playsInline
        className="fixed inset-0 w-full h-full object-cover"
      />

      {/* Dark overlay */}
      <div className="fixed inset-0 bg-black opacity-50" />

      {/*  Top bar with Logo and Name */}
      <header className="absolute top-6 left-6 flex items-center gap-3 z-10">
        <video
          src="/logo.mp4"
          autoPlay
          loop
          muted
          playsInline
          className="w-14 h-14 rounded-full object-cover shadow-md"
        />
        <h2 className="text-2xl font-bold text-white tracking-wide">
          Ink-Finity
        </h2>
      </header>

      {/* Foreground content */}
      <div className="relative z-10 container mx-auto px-4 py-16 text-center text-white space-y-16">
        {/* Hero Section */}
        <section className="flex flex-col-reverse lg:flex-row items-center gap-12">
          <div className="flex-1 space-y-6">
            <h1 className="text-5xl font-extrabold leading-tight">
              Give Life To Your Ideas
            </h1>
```

Figure 21: Front-End